SSLC • MATHEMATICS
Master Set • 2026 – 27

Dr. B R Ambedkar Residential School, Shankaranarayana

KARNATAKA STATE • KSEAB SYLLABUS

MATHEMATICS

Class X • SSLC • Subject Code 81E

FORMATIVE ASSESSMENT — MASTER SET

FA – 1 • FA – 2 • FA – 3 • FA – 4

Question Papers · Blue Print · Key Answers · Answer Booklet

Versions A • B • C • D

FA-1	Real Numbers · Polynomials · Linear Equations
FA-2	Quadratics · AP · Triangles · Coordinate Geometry
FA-3	Trigonometry · Heights · Circles · Areas
FA-4	Statistics · Surface Areas & Volumes · Probability

Each set : 45 Minutes • 20 Marks • Four parallel versions per Formative Assessment

Aligned to the current KSEAB syllabus • Compiled from official previous-year papers (2019 – 2026)

FORMATIVE ASSESSMENT

FA – 1

Real Numbers · Polynomials · Pair of Linear Equations in Two Variables

In this section : Blue Print • Question Papers (Versions A, B, C, D) • Key Answers • Answer Booklet

FA – 1 | BLUE PRINT

Class X • Chapters 1 – 3 • Time : 45 Minutes • Maximum Marks : 20

Chapter	MCQ 1 m	SA 1 m	2 m	3 m	4 m (Graph)	No. of Qns	Total Marks
1. Real Numbers	1 (Q1)	1 (Q4)	1 (Q5)	1 (Q8)	–	4	7
2. Polynomials	–	1 (Q3)	1 (Q6)	1 (Q9)	–	3	6
3. Pair of Linear Equations in Two Variables	1 (Q2)	–	1 (Q7)	–	1 (Q10)	3	7
Total	**2×1=2**	**2×1=2**	**3×2=6**	**2×3=6**	**1×4=4**	**10**	**20**

Weightage to Objectives

Objective	Questions	Marks	Percentage
Knowledge	Q1, Q2, Q3, Q4	4	20%
Understanding	Q5, Q6, Q9	8	40%
Application / Skill	Q7, Q8, Q10 (Graph)	8	40%
Total	**10 Questions**	**20**	**100%**

School : _____

FORMATIVE ASSESSMENT — 1

Class X • Mathematics (81E) • Version A

Chapters : Real Numbers · Polynomials · Pair of Linear Equations in Two Variables

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. If a and b are any two positive integers, then HCF(a, b) × LCM(a, b) is equal to .....
(A) a + b (B) a – b (C) a × b (D) a ÷ b

2. The lines represented by the equations 2x + 3y – 9 = 0 and 4x + 6y – 18 = 0 are .....
(A) intersecting lines (B) perpendicular lines (C) parallel lines (D) coincident lines

II. Answer the following questions. 2 × 1 = 2

3. Write the degree of the polynomial $p(x) = x(x^2 + 3) + 5x^2 + 7$.

4. Express 70 as a product of its prime factors.

III. Answer the following questions. 3 × 2 = 6

5. Prove that $3 + \sqrt{2}$ is an irrational number.

6. The sum and product of the zeroes of a quadratic polynomial $p(x) = ax^2 + bx + c$ are –3 and 2 respectively. Show that b + c = 5a.

7. Solve the given pair of linear equations by the elimination method :
2x + y = 8 ; x – y = 1

IV. Answer the following questions. 2 × 3 = 6

8. Find the LCM and HCF of the numbers 510 and 92 by the prime factorisation method and also verify that the product of the LCM and HCF of those numbers is equal to the product of the numbers.

9. Find the zeroes of the quadratic polynomial $p(x) = x^2 + 7x + 10$ and verify the relationship between the zeroes and the coefficients.

V. Answer the following question. 1 × 4 = 4

10. Find the solution of the given pair of linear equations by the graphical method :
x + 2y = 6 ; x + y = 5

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School : _____

FORMATIVE ASSESSMENT — 1

Class X • Mathematics (81E) • Version B

Chapters : Real Numbers · Polynomials · Pair of Linear Equations in Two Variables

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The product of the HCF and LCM of the two numbers 15 and 20 is .....
(A) 15 (B) 20 (C) 300 (D) 35

2. The lines represented by the equations $4x + 5y - 10 = 0$ and $8x + 10y + 20 = 0$ are .....
(A) intersecting lines (B) perpendicular lines (C) coincident lines (D) parallel lines

II. Answer the following questions. 2 × 1 = 2

3. Write the degree of the polynomial $p(x) = 3x^3 - x^4 + 2x^2 + 5x + 2$.

4. Express 96 as a product of its prime factors.

III. Answer the following questions. 3 × 2 = 6

5. Prove that $5 + \sqrt{3}$ is an irrational number.

6. If α and β are the zeroes of the polynomial $p(x) = 3x^2 - 12x + 15$, find the value of $\alpha^2 + \beta^2$.

7. Solve the given pair of linear equations by the elimination method :
$x + 2y = 5$; $x - y = 2$

IV. Answer the following questions. 2 × 3 = 6

8. Find the LCM and HCF of the numbers 72 and 120 by the prime factorisation method and verify that the product of the LCM and HCF is equal to the product of the two numbers.

9. Find the zeroes of the quadratic polynomial $p(x) = x^2 - 5x + 6$ and verify the relationship between the zeroes and the coefficients.

V. Answer the following question. 1 × 4 = 4

10. Find the solution of the given pair of linear equations by the graphical method :
$x + y = 7$; $3x - y = 1$

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School : _____

FORMATIVE ASSESSMENT — 1

Class X • Mathematics (81E) • Version C

Chapters : Real Numbers · Polynomials · Pair of Linear Equations in Two Variables

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. 7 × 11 × 13 + 13 is a .....
(A) prime number (B) composite number (C) irrational number (D) odd prime

2. The lines represented by the equations x – 2y = 0 and 3x + 4y – 20 = 0 are .....
(A) intersecting lines (B) parallel lines (C) coincident lines (D) perpendicular lines

II. Answer the following questions. 2 × 1 = 2

3. Write the degree of the polynomial $p(x) = 5x^2 - 6x^3 - 7x + 1$.

4. If $200 = 2^m \times 5^n$, then find the values of m and n.

III. Answer the following questions. 3 × 2 = 6

5. Prove that $2 + \sqrt{3}$ is an irrational number.

6. Find the value of k of the polynomial $p(x) = 2x^2 - 6x + k$, such that the sum of its zeroes is equal to half of the product of its zeroes.

7. Solve the given pair of linear equations by the elimination method :
$2x + y = 3$; $4x - y = 9$

IV. Answer the following questions. 2 × 3 = 6

8. Find the LCM and HCF of the numbers 36 and 48 by the prime factorisation method and verify that the product of the LCM and HCF is equal to the product of the two numbers.

9. Find the zeroes of the quadratic polynomial $p(x) = x^2 - 2x - 8$ and verify the relationship between the zeroes and the coefficients.

V. Answer the following question. 1 × 4 = 4

10. Find the solution of the given pair of linear equations by the graphical method :
$x + y = 5$; $2x + y = 7$

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School : _____

FORMATIVE ASSESSMENT — 1

Class X • Mathematics (81E) • Version D

Chapters : Real Numbers · Polynomials · Pair of Linear Equations in Two Variables

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The HCF of the smallest prime number and the smallest composite number is .....
(A) 1 (B) 2 (C) 4 (D) 8

2. The pair of linear equations $x + 2y = 4$ and $2x + 4y = 12$ has .....
(A) a unique solution (B) exactly two solutions (C) infinitely many solutions (D) no solution

II. Answer the following questions. 2 × 1 = 2

3. If $p(x) = x^2 - 3x + 2$, then find the value of $p(2)$.

4. Express 120 as a product of its prime factors.

III. Answer the following questions. 3 × 2 = 6

5. Prove that $3 + 2\sqrt{5}$ is an irrational number.

6. Find a quadratic polynomial whose sum and product of its zeroes are 6 and 8 respectively.

7. Solve the given pair of linear equations by the elimination method :
$x + y = 10$; $x - y = 4$

IV. Answer the following questions. 2 × 3 = 6

8. Find the LCM and HCF of the numbers 24 and 90 by the prime factorisation method and verify that the product of the LCM and HCF is equal to the product of the two numbers.

9. Find the zeroes of the quadratic polynomial $p(x) = x^2 - 7x + 12$ and verify the relationship between the zeroes and the coefficients.

V. Answer the following question. 1 × 4 = 4

10. Find the solution of the given pair of linear equations by the graphical method :
$x + y = 6$; $x - y = 2$

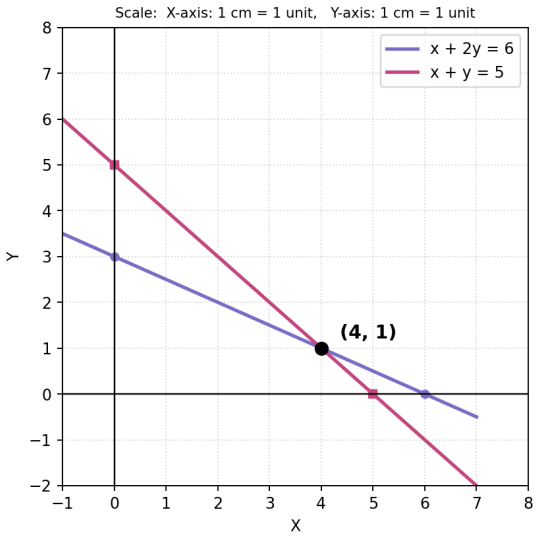
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FA – 1 | KEY ANSWERS • Version A

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(C) a × b	1
2	(D) coincident lines [$a_1/a_2 = b_1/b_2 = c_1/c_2$: 2/4 = 3/6 = 9/18 = ½]	1
3	$p(x) = x^3 + 3x + 5x^2 + 7$; Degree of the polynomial = 3	1
4	70 = 2 × 5 × 7	1
5	Let 3 + √2 be rational. Then 3 + √2 = a/b, where a, b are co-primes, b ≠ 0. (½) ⇒ √2 = (a – 3b)/b. (½) Since a, b are integers, (a – 3b)/b is rational ⇒ √2 is rational. (½) This contradicts that √2 is irrational. ∴ 3 + √2 is an irrational number. (½)	2
6	Sum of zeroes : α + β = –b/a = –3 ⇒ b = 3a (1) Product of zeroes : αβ = c/a = 2 ⇒ c = 2a (½) ∴ b + c = 3a + 2a = 5a (½)	2
7	2x + y = 8 ...(1) ; x – y = 1 ...(2) Adding (1) and (2) : 3x = 9 ⇒ x = 3 (1) Substituting in (1) : 6 + y = 8 ⇒ y = 2 ∴ x = 3, y = 2 (1)	2
8	510 = 2 × 3 × 5 × 17 ; 92 = 2^2 × 23 (1) HCF = 2 ; LCM = 2^2 × 3 × 5 × 17 × 23 = 23460 (1) Verify : LCM × HCF = 23460 × 2 = 46920 ; product = 510 × 92 = 46920 ∴ equal. (1)	3
9	$p(x) = x^2 + 7x + 10 = (x + 5)(x + 2)$ (1) ∴ zeroes are –5 and –2 (1) Sum = –7 = –b/a ; Product = 10 = c/a (1)	3
10	x + 2y = 6 → x : 0, 6, 2 ; y : 3, 0, 2 x + y = 5 → x : 0, 5, 3 ; y : 5, 0, 2 Table : 1 + 1 ; two lines drawn correctly : 1 ; reading the solution : 1 From the graph the lines intersect at (4, 1). ∴ x = 4, y = 1	4

Graph for Q.10

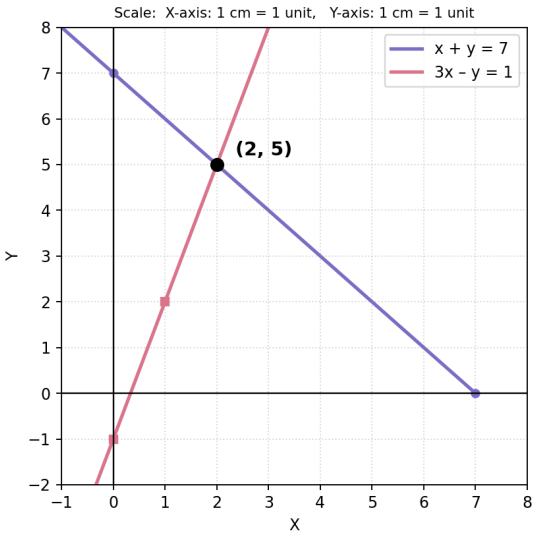


FA – 1 | KEY ANSWERS • Version B

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(C) 300 [HCF × LCM = product of the numbers = 15 × 20 = 300]	1
2	(D) parallel lines [$a_1/a_2 = b_1/b_2 \neq c_1/c_2$: 4/8 = 5/10 ≠ –10/20]	1
3	Degree of the polynomial = 4	1
4	$96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3 = 2^5 \times 3$	1
5	Let 5 + √3 be rational. Then 5 + √3 = a/b, where a, b are co-primes, b ≠ 0. (½) ⇒ √3 = (a – 5b)/b. (½) Since a, b are integers, (a – 5b)/b is rational ⇒ √3 is rational. (½) This contradicts that √3 is irrational. ∴ 5 + √3 is irrational. (½)	2
6	α + β = –b/a = 12/3 = 4 ; αβ = c/a = 15/3 = 5 (1) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (4)^2 - 2(5) = 16 - 10 = 6$ (1)	2
7	x + 2y = 5 ...(1) ; x – y = 2 ...(2) Subtracting (2) from (1) : 3y = 3 ⇒ y = 1 (1) Substituting in (2) : x = 2 + 1 = 3 ∴ x = 3, y = 1 (1)	2
8	$72 = 2^3 \times 3^2$; $120 = 2^3 \times 3 \times 5$ (1) HCF = $2^3 \times 3 = 24$; LCM = $2^3 \times 3^2 \times 5 = 360$ (1) Verify : LCM × HCF = 360 × 24 = 8640 ; product = 72 × 120 = 8640 ∴ equal. (1)	3
9	$p(x) = x^2 - 5x + 6 = x^2 - 3x - 2x + 6 = (x - 3)(x - 2)$ (1) ∴ zeroes are 3 and 2 (1) Sum = 3 + 2 = 5 = –b/a ; Product = 3 × 2 = 6 = c/a (1)	3
10	x + y = 7 → x : 0, 7, 2 ; y : 7, 0, 5 3x – y = 1 → x : 0, 1, 2 ; y : –1, 2, 5 Table : 1 + 1 ; two lines drawn correctly : 1 ; reading the solution : 1 From the graph the lines intersect at (2, 5). ∴ x = 2, y = 5	4

Graph for Q.10

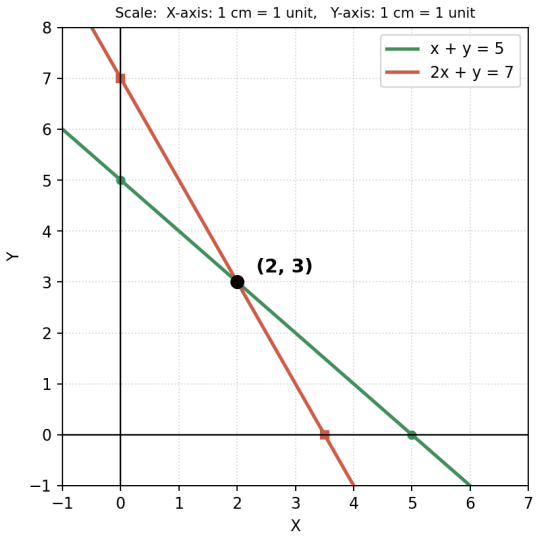


FA – 1 | KEY ANSWERS • Version C

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(B) composite number [$7 \times 11 \times 13 + 13 = 13(7 \times 11 + 1) = 13 \times 78$, which has factors other than 1 and itself]	1
2	(A) intersecting lines [$a_1/a_2 = 1/3$, $b_1/b_2 = -2/4$; $a_1/a_2 \neq b_1/b_2$]	1
3	Degree of the polynomial = 3	1
4	$200 = 2 \times 2 \times 2 \times 5 \times 5 = 2^3 \times 5^2$ ∴ m = 3 and n = 2	1
5	Let $2 + \sqrt{3}$ be rational. Then $2 + \sqrt{3} = a/b$, where a, b are co-primes, $b \neq 0$. (½) $\Rightarrow \sqrt{3} = (a - 2b)/b$. (½) Since a, b are integers, $(a - 2b)/b$ is rational $\Rightarrow \sqrt{3}$ is rational. (½) This contradicts that $\sqrt{3}$ is irrational. ∴ $2 + \sqrt{3}$ is an irrational number. (½)	2
6	Sum of zeroes = $-b/a = 6/2 = 3$; Product of zeroes = $c/a = k/2$ (1) Given : sum = ½ × product $\Rightarrow 3 = ½ \times (k/2) = k/4 \Rightarrow k = 12$ (1)	2
7	$2x + y = 3$ ...(1) ; $4x - y = 9$ ...(2) Adding (1) and (2) : $6x = 12 \Rightarrow x = 2$ (1) Substituting in (1) : $4 + y = 3 \Rightarrow y = -1$ ∴ x = 2, y = –1 (1)	2
8	$36 = 2^2 \times 3^2$; $48 = 2^4 \times 3$ (1) HCF = $2^2 \times 3 = 12$; LCM = $2^4 \times 3^2 = 144$ (1) Verify : LCM × HCF = 144 × 12 = 1728 ; product = 36 × 48 = 1728 ∴ equal. (1)	3
9	$p(x) = x^2 - 2x - 8 = x^2 - 4x + 2x - 8 = (x - 4)(x + 2)$ (1) ∴ zeroes are 4 and –2 (1) Sum = 4 + (–2) = 2 = –b/a ; Product = 4 × (–2) = –8 = c/a (1)	3
10	$x + y = 5 \rightarrow$ x : 0, 5, 2 ; y : 5, 0, 3 $2x + y = 7 \rightarrow$ x : 0, 3·5, 2 ; y : 7, 0, 3 Table : 1 + 1 ; two lines drawn correctly : 1 ; reading the solution : 1 From the graph the lines intersect at (2, 3). ∴ x = 2, y = 3	4

Graph for Q.10

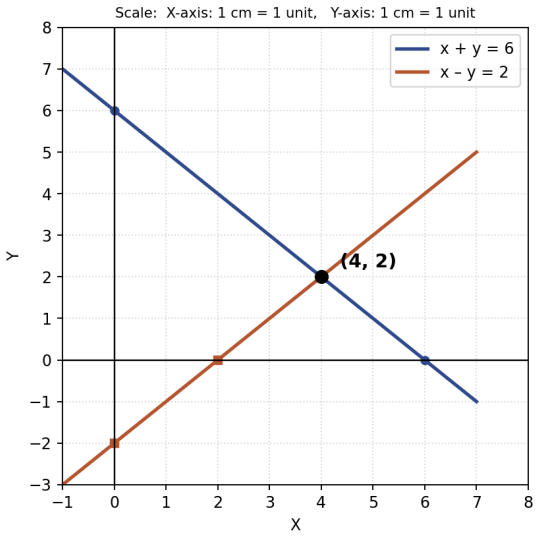


FA – 1 | KEY ANSWERS • Version D

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(B) 2 [smallest prime = 2, smallest composite = 4 ; HCF(2, 4) = 2]	1
2	(D) no solution [$a_1/a_2 = 1/2 = b_1/b_2 = 2/4$, but $\neq c_1/c_2 = 4/12 \Rightarrow$ parallel lines]	1
3	$p(2) = (2)^2 - 3(2) + 2 = 4 - 6 + 2 = 0$	1
4	$120 = 2 \times 2 \times 2 \times 3 \times 5 = 2^3 \times 3 \times 5$	1
5	Let $3 + 2\sqrt{5}$ be rational. Then $3 + 2\sqrt{5} = a/b$, where a, b are co-primes, $b \neq 0$. (½) $\Rightarrow 2\sqrt{5} = (a - 3b)/b \Rightarrow \sqrt{5} = (a - 3b)/(2b)$. (½) Since a, b are integers, the RHS is rational $\Rightarrow \sqrt{5}$ is rational. (½) This contradicts that $\sqrt{5}$ is irrational. $\therefore 3 + 2\sqrt{5}$ is an irrational number. (½)	2
6	Sum of zeroes = 6 ; Product of zeroes = 8 (1) Quadratic polynomial = x^2 – (sum)x + product = $x^2 - 6x + 8$ (1)	2
7	$x + y = 10$ ...(1) ; $x - y = 4$ ...(2) Adding (1) and (2) : $2x = 14 \Rightarrow x = 7$ (1) From (1) : $y = 10 - 7 = 3 \therefore x = 7, y = 3$ (1)	2
8	$24 = 2^3 \times 3$; $90 = 2 \times 3^2 \times 5$ (1) HCF $= 2 \times 3 = 6$; LCM $= 2^3 \times 3^2 \times 5 = 360$ (1) Verify : LCM × HCF $= 360 \times 6 = 2160$; product $= 24 \times 90 = 2160 \therefore$ equal. (1)	3
9	$p(x) = x^2 - 7x + 12 = x^2 - 3x - 4x + 12 = (x - 3)(x - 4)$ (1) $\therefore$ zeroes are 3 and 4 (1) Sum $= 3 + 4 = 7 = -b/a$; Product $= 3 \times 4 = 12 = c/a$ (1)	3
10	$x + y = 6 \rightarrow x$: 0, 6, 4 ; y : 6, 0, 2 $x - y = 2 \rightarrow x$: 2, 0, 4 ; y : 0, –2, 2 Table : 1 + 1 ; two lines drawn correctly : 1 ; reading the solution : 1 From the graph the lines intersect at (4, 2). $\therefore x = 4, y = 2$	4

Graph for Q.10

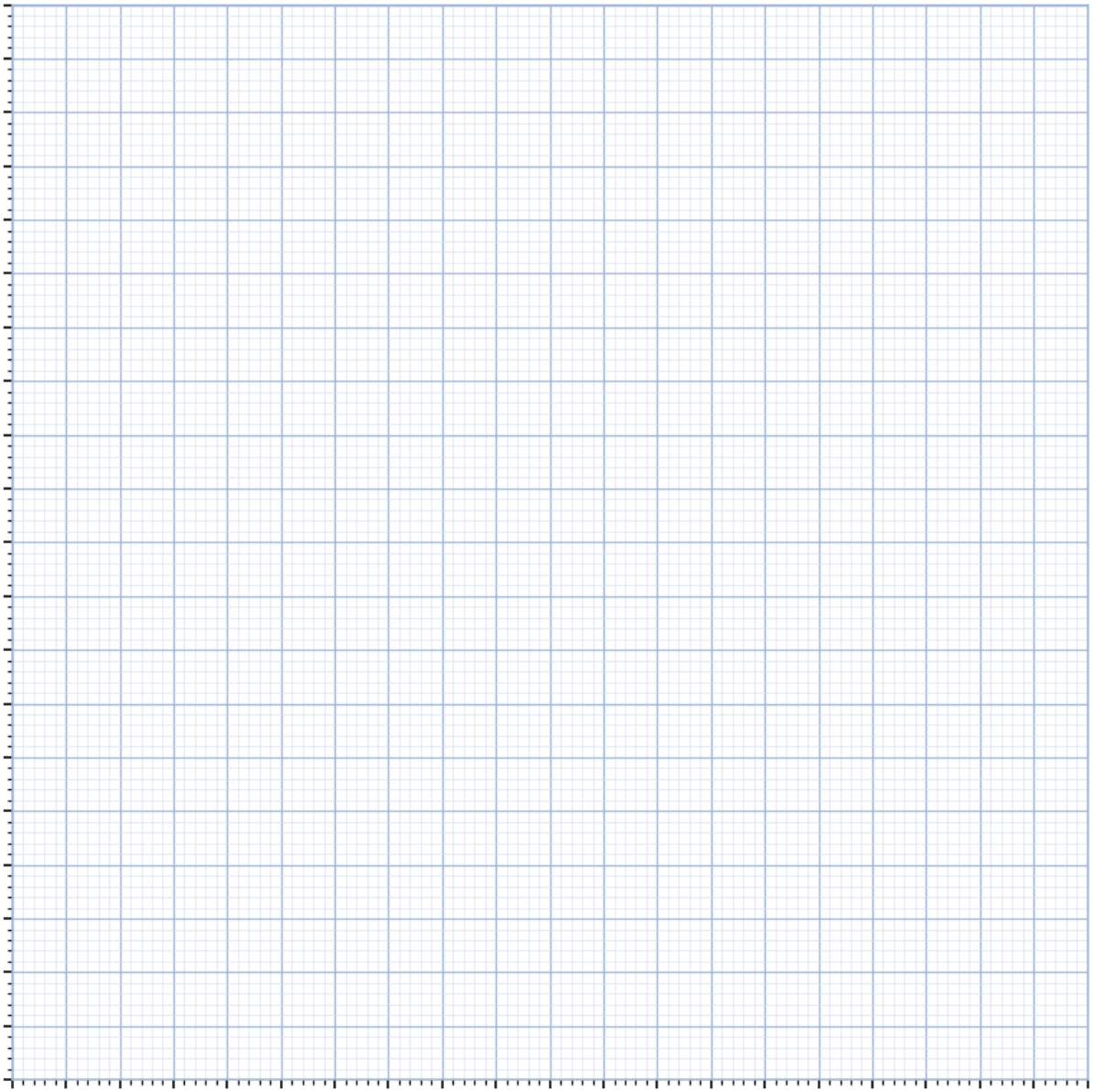


FA – 1 | ANSWER BOOKLET

Name : _____	**Roll No. : _____**
Marks Obtained : _____ / 20	**Invigilator's Signature : _____**

Q.No	Space for Answers

FA – 1 | GRAPH SHEET



FORMATIVE ASSESSMENT

FA – 2

Quadratic Equations · Arithmetic Progressions · Triangles · Coordinate Geometry

In this section : Blue Print • Question Papers (Versions A, B, C, D) • Key Answers • Answer Booklet

FA – 2 | BLUE PRINT

Class X • Chapters 4 – 7 • Time : 45 Minutes • Maximum Marks : 20

Chapter	MCQ 1 m	SA 1 m	2 m	3 m	4 m (Theorem)	No. of Qns	Total Marks
4. Quadratic Equations	1 (Q1)	–	1 (Q5)	1 (Q8)	–	3	6
5. Arithmetic Progressions	1 (Q2)	–	1 (Q6)	1 (Q9)	–	3	6
6. Triangles	–	1 (Q3)	–	–	1 (Q10)	2	5
7. Coordinate Geometry	–	1 (Q4)	1 (Q7)	–	–	2	3
Total	**2×1=2**	**2×1=2**	**3×2=6**	**2×3=6**	**1×4=4**	**10**	**20**

Weightage to Objectives

Objective	Questions	Marks	Percentage
Knowledge	Q1, Q2, Q3, Q4	4	20%
Understanding	Q5, Q6, Q7	6	30%
Application / Skill	Q8, Q9, Q10 (Theorem)	10	50%
Total	**10 Questions**	**20**	**100%**

School : _____

FORMATIVE ASSESSMENT — 2

Class X • Mathematics (81E) • Version A

Chapters : Quadratic Equations · Arithmetic Progressions · Triangles · Coordinate Geometry

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The discriminant of the quadratic equation $ax^2 + bx + c = 0$ is .....
(A) $b^2 - 4ac$ (B) $c^2 - 4ab$ (C) $b^2 + 4ac$ (D) $a^2 + 4ab$

2. If the nth term of an Arithmetic Progression is $a_n = 3n + 1$, then its 4th term is .....
(A) 10 (B) 13 (C) 11 (D) 12

II. Answer the following questions. 2 × 1 = 2

3. State the Basic Proportionality Theorem.

4. Find the distance of the point M(4, 3) from the origin.

III. Answer the following questions. 3 × 2 = 6

5. Find the discriminant of the quadratic equation $2x^2 - 5x + 3 = 0$ and hence write the nature of its roots.

6. Find the 20th term of the Arithmetic Progression 4, 7, 10, ... using the formula.

7. Find the coordinates of the point which divides the line segment joining the points (1, 6) and (4, 3) in the ratio 1 : 2 internally.

IV. Answer the following questions. 2 × 3 = 6

8. A train travels 360 km at a uniform speed. If the speed had been 5 km/h more, it would have taken 1 hour less for the same journey. Find the speed of the train.

9. The sum of the first 9 terms of an Arithmetic Progression is 144 and its 9th term is 28. Find the first term and the common difference.

V. Answer the following question. 1 × 4 = 4

10. Prove the following theorem : “If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio.” (Basic Proportionality Theorem)

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School : _____

FORMATIVE ASSESSMENT — 2

Class X • Mathematics (81E) • Version B

Chapters : Quadratic Equations · Arithmetic Progressions · Triangles · Coordinate Geometry

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The discriminant of the quadratic equation $px^2 + qx + r = 0$ is .....
(A) $q^2 - 4pr$ (B) $q^2 + 4pr$ (C) $p^2 - 4pr$ (D) $p^2 - 4qr$

2. The 20th term of the Arithmetic Progression 1, 5, 9, 13, ... is .....
(A) 76 (B) 77 (C) 78 (D) 80

II. Answer the following questions. 2 × 1 = 2

3. In ΔABC, D and E are points on the sides AB and AC respectively such that DE ∥ BC. If AD = 2 cm, DB = 3 cm, AE = x cm and EC = 6 cm, find the value of x using the Basic Proportionality Theorem.

4. Find the distance between the points (2, 3) and (4, 1).

III. Answer the following questions. 3 × 2 = 6

5. Find the value of the discriminant of the quadratic equation $x^2 + 3x + 2 = 0$ and hence write the nature of its roots.

6. Find the 21st term of the Arithmetic Progression 5, 9, 13, ... using the formula.

7. Find the coordinates of the point which divides the line segment joining the points (–1, 7) and (4, –3) in the ratio 2 : 3 internally.

IV. Answer the following questions. 2 × 3 = 6

8. The distance between two cities A and B is 132 km. Because of a flyover, the average speed of a car travelling on this route increases by 11 km/h and the car takes 1 hour less to cover the same distance. Find the original average speed of the car.

9. In an Arithmetic Progression, the sum of the first n terms is 210 and the sum of the first (n – 1) terms is 171. If the first term is 3, find the Arithmetic Progression and its 20th term.

V. Answer the following question. 1 × 4 = 4

10. Prove the following theorem : “If in two triangles the corresponding angles are equal, then their corresponding sides are in the same ratio (or proportion) and hence the two triangles are similar.” (AAA similarity criterion)

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School : _____

FORMATIVE ASSESSMENT — 2

Class X • Mathematics (81E) • Version C

Chapters : Quadratic Equations · Arithmetic Progressions · Triangles · Coordinate Geometry

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The roots of the quadratic equation (x – 3)(x + 2) = 0 are .....
(A) –3, 2 (B) 3, –2 (C) –3, –2 (D) 3, 2

2. If 8, x, 20 are in Arithmetic Progression, then the value of x is .....
(A) 12 (B) 14 (C) 16 (D) 28

II. Answer the following questions. 2 × 1 = 2

3. In ΔABC, D and E are points on the sides AB and AC respectively such that DE ∥ BC. If AD = 4 cm, DB = x cm, AE = 6 cm and EC = 9 cm, find the value of x using the Basic Proportionality Theorem.

4. Find the coordinates of the mid-point of the line segment joining the points (6, 3) and (4, 7).

III. Answer the following questions. 3 × 2 = 6

5. Find the value of the discriminant of the quadratic equation $x^2 + 4x + 4 = 0$ and hence write the nature of its roots.

6. Find the 30th term of the Arithmetic Progression 5, 8, 11, ... using the formula.

7. Find the coordinates of the point which divides the line segment joining the points (4, –3) and (8, 5) in the ratio 3 : 1 internally.

IV. Answer the following questions. 2 × 3 = 6

8. Find two consecutive positive integers, the sum of whose squares is 365.

9. In an Arithmetic Progression, the sum of the 4th and 8th terms is 24 and the sum of the 6th and 10th terms is 44. Find the first three terms of the Arithmetic Progression.

V. Answer the following question. 1 × 4 = 4

10. Prove the following theorem : “If one angle of a triangle is equal to one angle of another triangle and the sides including these angles are proportional, then the two triangles are similar.” (SAS similarity criterion)

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School : _____

FORMATIVE ASSESSMENT — 2

Class X • Mathematics (81E) • Version D

Chapters : Quadratic Equations · Arithmetic Progressions · Triangles · Coordinate Geometry

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. If one root of the quadratic equation $x^2 - 5x + 6 = 0$ is 2, then the other root is .....
(A) 1 (B) 3 (C) –2 (D) 6

2. The common difference of the Arithmetic Progression 100, 93, 86, ... is .....
(A) 7 (B) –7 (C) 93 (D) –93

II. Answer the following questions. 2 × 1 = 2

3. In ΔABC, D and E are points on the sides AB and AC respectively such that DE ∥ BC. If AD = 5 cm, DB = 3 cm, AE = x cm and EC = 6 cm, find the value of x using the Basic Proportionality Theorem.

4. Find the distance between the points (3, 4) and (6, 8).

III. Answer the following questions. 3 × 2 = 6

5. Find the value of the discriminant of the quadratic equation $3x^2 - 2x + 1 = 0$ and hence write the nature of its roots.

6. Find the 25th term of the Arithmetic Progression 3, 7, 11, ... using the formula.

7. Find the coordinates of the point which divides the line segment joining the points (2, 3) and (6, 7) in the ratio 1 : 3 internally.

IV. Answer the following questions. 2 × 3 = 6

8. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the lengths of the other two sides.

9. How many terms of the Arithmetic Progression 24, 21, 18, ... must be taken so that their sum is 78?

V. Answer the following question. 1 × 4 = 4

10. Prove the following theorem : “If in two triangles the corresponding angles are equal, then their corresponding sides are in the same ratio (or proportion) and hence the two triangles are similar.” (AAA similarity criterion)

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FA – 2 | KEY ANSWERS • Version A

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(A) $b^2 - 4ac$	1
2	(B) 13 [$a_n = 3(4) + 1 = 13$]	1
3	Statement : If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio.	1
4	$d = \sqrt{(4^2 + 3^2)} = \sqrt{(16 + 9)} = \sqrt{25} = 5$ units	1
5	$a = 2, b = -5, c = 3$; $D = b^2 - 4ac = 25 - 24 = 1$ (1) $D > 0$ and a perfect square ∴ roots are real, rational and distinct. (1)	2
6	$a = 4, d = 3$ (½) $a_{20} = a + 19d = 4 + 19 \times 3 = 4 + 57 = 61$ (1½)	2
7	$(x_1,y_1)=(1,6)$, $(x_2,y_2)=(4,3)$, $m_1:m_2 = 1:2$ (½) $x = (1\times4 + 2\times1)/3 = 6/3 = 2$ (½) $y = (1\times3 + 2\times6)/3 = 15/3 = 5$ (½) ∴ point of division = (2, 5) (½)	2
8	Let speed of train = x km/h. (½) $360/x - 360/(x+5) = 1 \Rightarrow 360(x+5) - 360x = x(x+5)$ (1) $1800 = x^2 + 5x \Rightarrow x^2 + 5x - 1800 = 0$ (½) $(x + 45)(x - 40) = 0 \Rightarrow x = -45$ or $x = 40$ (½) Speed cannot be negative ∴ speed of the train = 40 km/h (½)	3
9	$a_9 = a + 8d = 28$ ...(1) (½) $S_9 = (9/2)(2a + 8d) = 144 \Rightarrow a + 4d = 16$ ...(2) (1) (1) – (2) : $4d = 12 \Rightarrow d = 3$ (½) $a + 12 = 16 \Rightarrow a = 4$ ∴ first term = 4, common difference = 3 (1)	3
10	Statement : If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. (1) Given : In ΔABC, DE ∥ BC, D on AB, E on AC. To Prove : AD/DB = AE/EC. Construction : Draw DM ⊥ AC and EN ⊥ AB ; join BE and CD. (½) Proof : ar(ΔADE)/ar(ΔBDE) = AD/DB ...(1) (½) ar(ΔADE)/ar(ΔCDE) = AE/EC ...(2) (½) ΔBDE and ΔCDE are on the same base DE and between the same parallels DE and BC ⇒ ar(ΔBDE) = ar(ΔCDE) ...(3) (½) From (1), (2) and (3) : AD/DB = AE/EC (Hence proved). (1)	4

FA – 2 | KEY ANSWERS • Version B

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(A) $q^2 - 4pr$	1
2	(B) 77 [a = 1, d = 4 ; a_{20} = a + 19d = 1 + 76 = 77]	1
3	By the Basic Proportionality Theorem, AD/DB = AE/EC ⇒ 2/3 = x/6 ⇒ x = (2 × 6)/3 = 4 cm	1
4	d = √[(4 – 2)^2 + (1 – 3)^2] = √(4 + 4) = √8 = 2√2 units	1
5	a = 1, b = 3, c = 2 ; D = $b^2 - 4ac$ = 9 – 8 = 1 (1) D > 0 and a perfect square ∴ roots are real, rational and distinct. (1)	2
6	a = 5, d = 4 (½) a_{21} = a + 20d = 5 + 20 × 4 = 5 + 80 = 85 (1½)	2
7	$(x_1,y_1)=(-1,7)$, $(x_2,y_2)=(4,-3)$, $m_1:m_2$ = 2:3 (½) x = (2×4 + 3×–1)/5 = (8 – 3)/5 = 1 (½) y = (2×–3 + 3×7)/5 = (–6 + 21)/5 = 3 (½) ∴ point of division = (1, 3) (½)	2
8	Let original speed = x km/h. (½) 132/x – 132/(x+11) = 1 ⇒ 132(x+11) – 132x = x(x+11) (1) 1452 = x^2 + 11x ⇒ x^2 + 11x – 1452 = 0 (½) (x + 44)(x – 33) = 0 ⇒ x = –44 or x = 33 (½) Speed cannot be negative ∴ original speed = 33 km/h (½)	3
9	nth term $a_n = S_n - S_{n-1}$ = 210 – 171 = 39 (1) S_n = n/2 (a + a_n) ⇒ 210 = n/2 (3 + 39) = 21n ⇒ n = 10 (1) a + (n – 1)d = 39 ⇒ 3 + 9d = 39 ⇒ d = 4 ; AP : 3, 7, 11, 15, … a_{20} = 3 + 19 × 4 = 79 (1)	3
10	Statement : If in two triangles the corresponding angles are equal, then their corresponding sides are in the same ratio and the triangles are similar. (½) Given : In ΔABC and ΔDEF, ∠A = ∠D, ∠B = ∠E, ∠C = ∠F. To Prove : AB/DE = BC/EF = AC/DF. Construction : Mark P on DE and Q on DF such that DP = AB and DQ = AC. Join PQ. (½) Proof : In ΔABC and ΔDPQ : AB = DP, AC = DQ and ∠A = ∠D ⇒ ΔABC ≅ ΔDPQ (SAS). (1) ∴ ∠B = ∠DPQ ; but ∠B = ∠E ⇒ ∠DPQ = ∠E ⇒ PQ ∥ EF. (1) By the Basic Proportionality Theorem in ΔDEF : DP/DE = DQ/DF ⇒ AB/DE = AC/DF. Similarly AB/DE = BC/EF. ∴ AB/DE = BC/EF = AC/DF (Hence proved). (1)	4

FA – 2 | KEY ANSWERS • Version C

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(B) 3, –2	1
2	(B) 14 [x is the arithmetic mean : x = (8 + 20)/2 = 14]	1
3	By the Basic Proportionality Theorem, AD/DB = AE/EC ⇒ 4/x = 6/9 ⇒ 6x = 36 ⇒ x = 6 cm	1
4	Mid-point = ((6 + 4)/2 , (3 + 7)/2) = (5, 5)	1
5	a = 1, b = 4, c = 4 ; D = b^2 – 4ac = 16 – 16 = 0 (1) D = 0 ∴ the roots are real and equal. (1)	2
6	a = 5, d = 3 (½) a_{30} = a + 29d = 5 + 29 × 3 = 5 + 87 = 92 (1½)	2
7	(x_1,y_1)=(4,–3), (x_2,y_2)=(8,5), m_1:m_2 = 3:1 (½) x = (3×8 + 1×4)/4 = (24 + 4)/4 = 7 (½) y = (3×5 + 1×–3)/4 = (15 – 3)/4 = 3 (½) ∴ point of division = (7, 3) (½)	2
8	Let the integers be x and (x + 1). (½) $x^2 + (x + 1)^2 = 365 ⇒ 2x^2 + 2x + 1 = 365$ (1) $2x^2 + 2x – 364 = 0 ⇒ x^2 + x – 182 = 0$ (½) (x – 13)(x + 14) = 0 ⇒ x = 13 (positive) (½) ∴ the integers are 13 and 14. (½)	3
9	$a_4 + a_8 = 24 ⇒ (a + 3d) + (a + 7d) = 2a + 10d = 24 ⇒ a + 5d = 12$ …(1) (1) $a_6 + a_{10} = 44 ⇒ (a + 5d) + (a + 9d) = 2a + 14d = 44 ⇒ a + 7d = 22$ …(2) (1) (2) – (1) : 2d = 10 ⇒ d = 5 ; a = 12 – 25 = –13 ∴ first three terms : –13, –8, –3 (1)	3
10	Statement : If one angle of a triangle equals one angle of another and the sides including these angles are proportional, the triangles are similar. (½) Given : In ΔABC and ΔDEF, ∠A = ∠D and AB/DE = AC/DF. To Prove : ΔABC ~ ΔDEF. Construction : Mark P on DE and Q on DF such that DP = AB and DQ = AC. Join PQ. (½) Proof : In ΔABC and ΔDPQ : AB = DP, AC = DQ and ∠A = ∠D ⇒ ΔABC ≅ ΔDPQ (SAS). (1) From AB/DE = AC/DF : DP/DE = DQ/DF ⇒ by the converse of BPT, PQ ∥ EF. (1) ∴ ∠DPQ = ∠E and ∠DQP = ∠F (corresponding angles) ⇒ ∠B = ∠E, ∠C = ∠F. With ∠A = ∠D, by AAA similarity ΔABC ~ ΔDEF (Hence proved). (1)	4

FA – 2 | KEY ANSWERS • Version D

Class X • Mathematics (81E) • Maximum Marks : 20

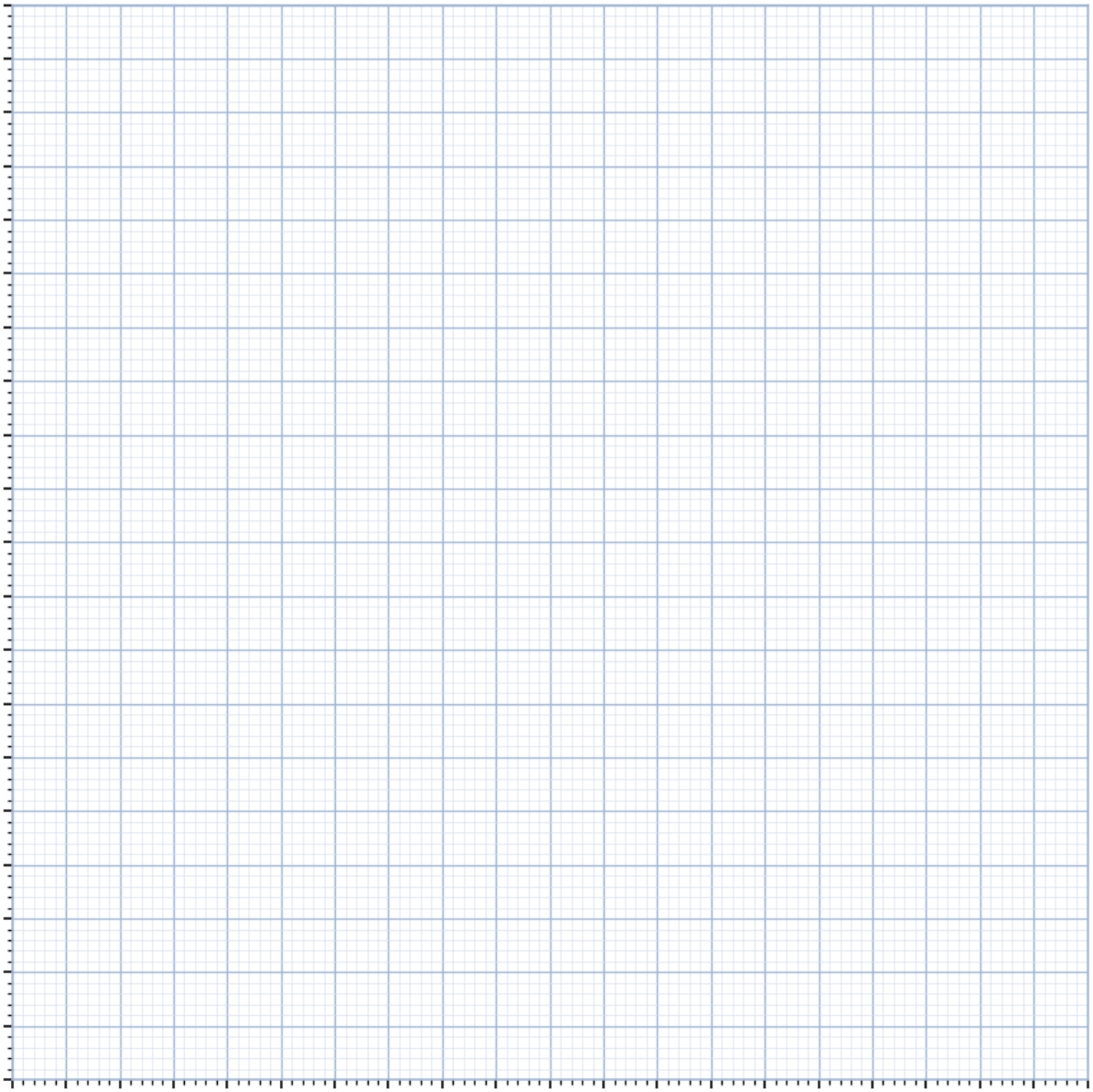
Q.	Value Points	Marks
1	(B) 3 [$x^2 - 5x + 6 = (x - 2)(x - 3) = 0 \Rightarrow$ roots 2 and 3 ; other root = 3]	1
2	(B) –7 [$d = 93 - 100 = -7$]	1
3	By the Basic Proportionality Theorem, AD/DB = AE/EC ⇒ 5/3 = x/6 ⇒ 3x = 30 ⇒ x = 10 cm	1
4	$d = \sqrt{[(6 - 3)^2 + (8 - 4)^2]} = \sqrt{(9 + 16)} = \sqrt{25} = 5$ units	1
5	a = 3, b = –2, c = 1 ; $D = b^2 - 4ac = 4 - 12 = -8$ (1) D < 0 ∴ the equation has no real roots. (1)	2
6	a = 3, d = 4 (½) $a_{25} = a + 24d = 3 + 24 \times 4 = 3 + 96 = 99$ (1½)	2
7	$(x_1,y_1)=(2,3)$, $(x_2,y_2)=(6,7)$, $m_1:m_2 = 1:3$ (½) $x = (1\times6 + 3\times2)/4 = (6 + 6)/4 = 3$ (½) $y = (1\times7 + 3\times3)/4 = (7 + 9)/4 = 4$ (½) ∴ point of division = (3, 4) (½)	2
8	Let base = x cm ; altitude = (x – 7) cm. (½) $x^2 + (x - 7)^2 = 13^2 \Rightarrow x^2 + x^2 - 14x + 49 = 169$ (1) $2x^2 - 14x - 120 = 0 \Rightarrow x^2 - 7x - 60 = 0$ (½) $(x - 12)(x + 5) = 0 \Rightarrow x = 12$ (½) ∴ base = 12 cm and altitude = 5 cm. (½)	3
9	$S_n = n/2\ [2a + (n - 1)d]$; a = 24, d = –3, $S_n = 78$ (½) $78 = n/2\ [48 + (n - 1)(-3)] = n/2\ [51 - 3n]$ (1) $156 = 51n - 3n^2 \Rightarrow 3n^2 - 51n + 156 = 0 \Rightarrow n^2 - 17n + 52 = 0$ (1) $(n - 4)(n - 13) = 0 \Rightarrow n = 4$ or $n = 13$ (½)	3
10	Statement : If in two triangles the corresponding angles are equal, then their corresponding sides are in the same ratio and the triangles are similar. (½) Given : In ΔABC and ΔDEF, ∠A = ∠D, ∠B = ∠E, ∠C = ∠F. To Prove : AB/DE = BC/EF = AC/DF. Construction : Mark P on DE and Q on DF such that DP = AB and DQ = AC. Join PQ. (½) Proof : In ΔABC and ΔDPQ : AB = DP, AC = DQ and ∠A = ∠D ⇒ ΔABC ≅ ΔDPQ (SAS). (1) ∴ ∠B = ∠DPQ ; but ∠B = ∠E ⇒ ∠DPQ = ∠E ⇒ PQ ∥ EF. (1) By the Basic Proportionality Theorem in ΔDEF : DP/DE = DQ/DF ⇒ AB/DE = AC/DF. Similarly AB/DE = BC/EF. ∴ AB/DE = BC/EF = AC/DF (Hence proved). (1)	4

FA – 2 | ANSWER BOOKLET

Name : _____	**Roll No. : _____**
Marks Obtained : _____ / 20	**Invigilator's Signature : _____**

Q.No	Space for Answers

FA – 2 | GRAPH SHEET



FORMATIVE ASSESSMENT

FA – 3

Introduction to Trigonometry · Applications of Trigonometry · Circles · Areas Related to Circles

In this section : Blue Print • Question Papers (Versions A, B, C, D) • Key Answers • Answer Booklet

FA – 3 | BLUE PRINT

Class X • Time : 45 Minutes • Maximum Marks : 20

Chapter	MCQ 1 m	SA 1 m	2 m	3 m	4 m	No. of Qns	Total Marks
Introduction to Trigonometry	1 (Q1)	–	1 (Q5)	1 (Q8)	–	3	6
Applications of Trigonometry	–	–	1 (Q7)	–	1 (Q10)	2	6
Circles	–	1 (Q4)	–	1 (Q9) Th.	–	2	4
Areas Related to Circles	1 (Q2)	1 (Q3)	1 (Q6)	–	–	3	4
Total	**2×1=2**	**2×1=2**	**3×2=6**	**2×3=6**	**1×4=4**	**10**	**20**

Weightage to Objectives

Objective	Questions	Marks	Percentage
Knowledge	Q1, Q2, Q3, Q4	4	20%
Understanding	Q5, Q6, Q8	7	35%
Application / Skill	Q7, Q9 (Theorem), Q10	9	45%
Total	**10 Questions**	**20**	**100%**

School : _____

FORMATIVE ASSESSMENT — 3

Class X • Mathematics (81E) • Version A

Chapters : Introduction to Trigonometry · Applications of Trigonometry · Circles · Areas Related to Circles

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The value of (sin 30° + cos 60° – tan 45°) is .....
(A) 1 (B) –1 (C) 2 (D) 0

2. If the area of a circle is 49π sq.units, then its perimeter is .....
(A) 7π units (B) 9π units (C) 14π units (D) 49π units

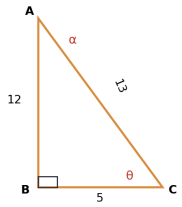
II. Answer the following questions. 2 × 1 = 2

3. Write the formula to find the area of a sector of a circle, if the angle at the centre is θ degrees.

4. How many tangents can be drawn to a circle from an external point?

III. Answer the following questions. 3 × 2 = 6

5. In the given figure, ∠ABC = 90°. Find the values of sin θ and tan α.



6. Find the area of a sector of a circle of radius 7 cm, if the angle at the centre is 90°. (Take π = 22/7)

7. The angle of elevation of the top of a vertical tower on a level ground from a point at a distance of 9√3 m from its foot is 60°. Find the height of the tower.

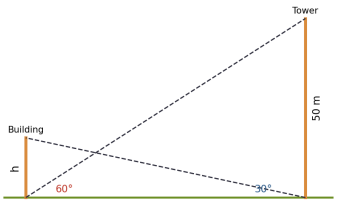
IV. Answer the following questions. 2 × 3 = 6

8. Prove that (sec A – cos A)(cot A + tan A) = tan A · sec A.

9. Prove that "the lengths of tangents drawn from an external point to a circle are equal."

V. Answer the following question. 1 × 4 = 4

10. A tower and a building are standing vertically on the same level ground. The angle of elevation of the top of the building from the foot of the tower is 30° and the angle of elevation of the top of the tower from the foot of the building is 60°. If the height of the tower is 50 m, find the height of the building.



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School : _____

FORMATIVE ASSESSMENT — 3

Class X • Mathematics (81E) • Version B

Chapters : Introduction to Trigonometry · Applications of Trigonometry · Circles · Areas Related to Circles

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The value of (cos 48° – sin 42°) is .....
(A) 0 (B) ¼ (C) ½ (D) 1

2. If the diameter of a circle is 14 cm, then its circumference is .....
(A) 28 cm (B) 44 cm (C) 56 cm (D) 88 cm

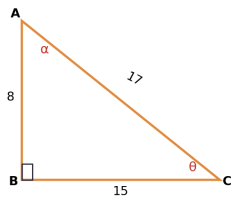
II. Answer the following questions. 2 × 1 = 2

3. Write the formula to find the length of an arc of a sector of a circle of radius r, when the angle at the centre is θ degrees.

4. What is the measure of the angle between a tangent to a circle and the radius drawn at the point of contact?

III. Answer the following questions. 3 × 2 = 6

5. In the given figure, ∠ABC = 90°. Find the values of sin θ and tan α.



6. Find the area of a sector of a circle of radius 21 cm, if the angle at the centre is 60°. (Take π = 22/7)

7. The angle of elevation of the top of a vertical tower from a point on the ground, 30 m away from its foot, is 30°. Find the height of the tower.

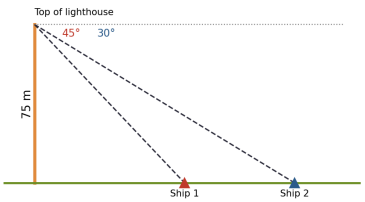
IV. Answer the following questions. 2 × 3 = 6

8. Prove that cosec A (1 – cos A)(cosec A + cot A) = 1.

9. Prove that "the tangent at any point of a circle is perpendicular to the radius through the point of contact."

V. Answer the following question. 1 × 4 = 4

10. As observed from the top of a 75 m high lighthouse from the sea-level, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships. (Take √3 = 1·73)



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School : _____

FORMATIVE ASSESSMENT — 3

Class X • Mathematics (81E) • Version C

Chapters : Introduction to Trigonometry · Applications of Trigonometry · Circles · Areas Related to Circles

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The value of $(\sec^2 26° - \tan^2 26°)$ is .....
(A) ½ (B) 0 (C) 2 (D) 1

2. The area of a circle of radius 7 cm is ($\pi = 22/7$) .....
(A) 154 cm^2 (B) 44 cm^2 (C) 22 cm^2 (D) 49 cm^2

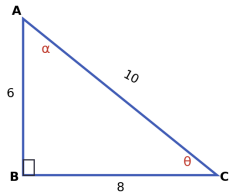
II. Answer the following questions. 2 × 1 = 2

3. If the perimeter and the area of a circle are numerically equal, then find the radius of the circle.

4. How many tangents can be drawn to a circle from a point lying on the circle?

III. Answer the following questions. 3 × 2 = 6

5. In the given figure, ∠ABC = 90°. Find the values of cos θ and tan α.



6. Find the area of a sector of a circle of radius 14 cm, if the angle at the centre is 45°. (Take π = 22/7)

7. The angle of elevation of the top of a vertical tower from a point on the ground, 25 m away from its foot, is 60°. Find the height of the tower. (Take √3 = 1·73)

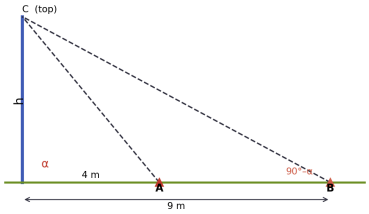
IV. Answer the following questions. 2 × 3 = 6

8. Prove that sec A (1 – sin A)(sec A + tan A) = 1.

9. Prove that "the lengths of tangents drawn from an external point to a circle are equal."

V. Answer the following question. 1 × 4 = 4

10. The angles of elevation of the top of a vertical tower from two points at distances of 4 m and 9 m from the base of the tower, in the same straight line with it, are complementary. Find the height of the tower.



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School : _____

FORMATIVE ASSESSMENT — 3

Class X • Mathematics (81E) • Version D

Chapters : Introduction to Trigonometry · Applications of Trigonometry · Circles · Areas Related to Circles

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 2 × 1 = 2

1. The value of $(\sin^2 35° + \cos^2 35°)$ is .....
(A) 0 (B) ½ (C) 1 (D) 2

2. The circumference of a circle of radius 14 cm is ($\pi = 22/7$) .....
(A) 44 cm (B) 88 cm (C) 22 cm (D) 154 cm

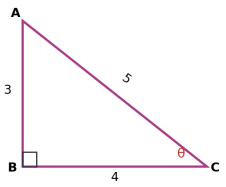
II. Answer the following questions. 2 × 1 = 2

3. Find the area of a quadrant of a circle of radius 7 cm. (Take $\pi = 22/7$)

4. What is the name given to a line which intersects a circle at two distinct points?

III. Answer the following questions. 3 × 2 = 6

5. In the given figure, $\angle ABC = 90°$. Find the values of sin θ and cos θ.



6. Find the length of the arc of a sector of a circle of radius 21 cm, if the angle at the centre is 120°. (Take $\pi = 22/7$)

7. From the top of a building 20 m high, the angle of depression of a car standing on the ground is 30°. Find the distance of the car from the foot of the building. (Take $\sqrt{3} = 1{\cdot}73$)

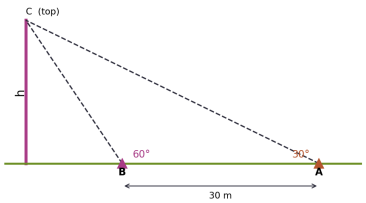
IV. Answer the following questions. 2 × 3 = 6

8. Prove that $\sin A / (1 + \cos A) + (1 + \cos A) / \sin A = 2 \operatorname{cosec} A$.

9. Prove that "the tangent at any point of a circle is perpendicular to the radius through the point of contact."

V. Answer the following question. 1 × 4 = 4

10. The angle of elevation of the top of a vertical tower from a point on the ground is 30°. On walking 30 m towards the tower, the angle of elevation becomes 60°. Find the height of the tower. (Take $\sqrt{3} = 1{\cdot}73$)



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FA – 3 | KEY ANSWERS • Version A

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(D) 0 [½ + ½ – 1 = 0]	1
2	(C) 14π units [$\pi r^2 = 49\pi \Rightarrow r = 7$; perimeter = $2\pi r = 14\pi$ units]	1
3	Area of a sector = $(\theta/360°) \times \pi r^2$	1
4	Two tangents can be drawn to a circle from an external point.	1
5	In right ΔABC, ∠B = 90°, AB = 12, BC = 5, AC = 13. sin θ = AB/AC = 12/13 (1) tan α = BC/AB = 5/12 (1)	2
6	Area of sector = $(\theta/360°) \times \pi r^2$ (½) = (90/360) × (22/7) × 7 × 7 (1) = (1/4) × 22 × 7 = 38·5 cm^2 (½)	2
7	tan 60° = h / 9√3 (1) √3 = h / 9√3 ⇒ h = 9√3 × √3 = 9 × 3 = 27 m (1)	2
8	LHS = (sec A – cos A)(cot A + tan A) = (1/cos A – cos A)(cos A/sin A + sin A/cos A) (½) = $[(1 - \cos^2 A)/\cos A] \times [(\cos^2 A + \sin^2 A)/(\sin A \cdot \cos A)]$ (1) = $(\sin^2 A/\cos A) \times (1/(\sin A \cdot \cos A))$ (½) = $\sin A/\cos^2 A$ = (sin A/cos A)(1/cos A) = tan A · sec A = RHS (Hence proved). (1)	3
9	Data : O is the centre ; PQ and PR are tangents from an external point P. (½) To Prove : PQ = PR. Construction : Join OP, OQ and OR. (½) Proof : In ΔOQP and ΔORP : ∠OQP = ∠ORP = 90° (radius ⊥ tangent) (½) OQ = OR (radii) ; OP = OP (common) (½) ∴ ΔOQP ≅ ΔORP (RHS congruence) (½) ∴ PQ = PR (CPCT) (Hence proved). (½)	3
10	In ΔCDB : tan 60° = CD/BD ⇒ √3 = 50/BD ⇒ BD = 50/√3 m (1) In ΔABD : tan 30° = AB/BD ⇒ 1/√3 = h/(50/√3) (1) h = (50/√3) × (1/√3) = 50/3 (1) ∴ height of the building = 50/3 ≈ 16·67 m (1)	4

FA – 3 | KEY ANSWERS • Version B

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(A) 0 [cos 48° = cos(90° – 42°) = sin 42° ∴ cos 48° – sin 42° = 0]	1
2	(B) 44 cm [circumference = πd = (22/7) × 14 = 44 cm]	1
3	Length of arc = (θ / 360°) × 2πr	1
4	The tangent and the radius at the point of contact are perpendicular, i.e. the angle is 90°.	1
5	Right ΔABC, ∠B = 90°, AB = 8, BC = 15, AC = 17. sin θ = AB / AC = 8/17 (1) tan α = BC / AB = 15/8 (1)	2
6	Area of sector = (θ/360°) × πr^2 (½) = (60/360) × (22/7) × 21 × 21 (1) = (1/6) × 22 × 3 × 21 = 231 cm^2 (½)	2
7	tan 30° = h / 30 (1) 1/√3 = h / 30 ⇒ h = 30/√3 = 10√3 m (1) (≈ 17·32 m)	2
8	LHS = cosec A (1 – cos A)(cosec A + cot A) = (1/sin A)(1 – cos A)(1/sin A + cos A/sin A) (1) = (1/sin A)(1 – cos A)·(1 + cos A)/sin A = (1 – $\cos^2$A)/$\sin^2$A (1) = $\sin^2$A / $\sin^2$A = 1 = RHS (Hence proved). (1)	3
9	Data : O is the centre ; XY is a tangent at point P ; OP is the radius. (½) To Prove : OP ⊥ XY. Construction : Take any point Q (other than P) on XY and join OQ. (½) Proof : Q lies outside the circle (every point of a tangent except the point of contact lies outside), so OQ > OP. (1) This is true for every point Q on XY. Hence OP is the shortest distance from O to XY. The shortest distance from a point to a line is the perpendicular. ∴ OP ⊥ XY (Hence proved). (1)	3
10	Let lighthouse CD = 75 m ; ships at A (45°) and B (30°), A nearer. In Δ with 45° : tan 45° = 75 / DA ⇒ DA = 75 m (1) In Δ with 30° : tan 30° = 75 / DB ⇒ 1/√3 = 75/DB ⇒ DB = 75√3 m (1½) Distance between ships AB = DB – DA = 75√3 – 75 = 75(√3 – 1) (1) = 75(1·73 – 1) = 75 × 0·73 = 54·75 m (½)	4

FA – 3 | KEY ANSWERS • Version C

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(D) 1 [$\sec^2\theta - \tan^2\theta = 1$ for all θ]	1
2	(A) 154 cm^2 [$\pi r^2 = (22/7) \times 7 \times 7 = 154\ cm^2$]	1
3	Perimeter = Area $\Rightarrow 2\pi r = \pi r^2 \Rightarrow r = 2$ units	1
4	Only one tangent can be drawn to a circle from a point lying on the circle.	1
5	Right ΔABC, $\angle B = 90°$, AB = 6, BC = 8, AC = 10. $\cos\theta = BC / AC = 8/10 = 4/5$ (1) $\tan\alpha = BC / AB = 8/6 = 4/3$ (1)	2
6	Area of sector = $(\theta/360°) \times \pi r^2$ (½) $= (45/360) \times (22/7) \times 14 \times 14$ (1) $= (1/8) \times 22 \times 28 = 77\ cm^2$ (½)	2
7	$\tan 60° = h / 25$ (1) $\sqrt{3} = h / 25 \Rightarrow h = 25\sqrt{3} = 25 \times 1{\cdot}73 = 43{\cdot}25$ m (1)	2
8	LHS = sec A (1 – sin A)(sec A + tan A) $= (1/\cos A)(1 - \sin A)(1/\cos A + \sin A/\cos A)$ (1) $= (1/\cos A)(1 - \sin A)\cdot(1 + \sin A)/\cos A = (1 - \sin^2 A)/\cos^2 A$ (1) $= \cos^2 A / \cos^2 A = 1$ = RHS (Hence proved). (1)	3
9	Data : O is the centre ; PQ and PR are tangents from an external point P. (½) To Prove : PQ = PR. Construction : Join OP, OQ and OR. (½) Proof : In ΔOQP and ΔORP : $\angle OQP = \angle ORP = 90°$ (radius $\perp$ tangent) (½) OQ = OR (radii) ; OP = OP (common) (½) $\therefore \Delta OQP \cong \Delta ORP$ (RHS congruence) (½) $\therefore$ PQ = PR (CPCT) (Hence proved). (½)	3
10	Let tower CD = h ; A and B are 4 m and 9 m from foot D, angles complementary. From A : $\tan\alpha = h / 4$; From B : $\tan(90° - \alpha) = h / 9 \Rightarrow \cot\alpha = h / 9$ (1) $\tan\alpha \times \cot\alpha = 1 \Rightarrow (h/4)(h/9) = 1$ (1) $h^2 / 36 = 1 \Rightarrow h^2 = 36$ (1) $\therefore$ height of the tower h = 6 m (1)	4

FA – 3 | KEY ANSWERS • Version D

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(C) 1 [$\sin^2\theta + \cos^2\theta = 1$ for all θ]	1
2	(B) 88 cm [circumference = $2\pi r = 2 \times (22/7) \times 14 = 88$ cm]	1
3	Area of a quadrant = $(1/4)\pi r^2 = (1/4) \times (22/7) \times 7 \times 7 = 38{\cdot}5$ cm^2	1
4	Such a line is called a secant of the circle.	1
5	Right ΔABC, ∠B = 90°, AB = 3, BC = 4, AC = 5. sin θ = AB / AC = 3/5 (1) cos θ = BC / AC = 4/5 (1)	2
6	Length of arc = (θ/360°) × 2πr (½) = (120/360) × 2 × (22/7) × 21 (1) = (1/3) × 2 × 22 × 3 = 44 cm (½)	2
7	Angle of depression = angle of elevation from the car = 30°. tan 30° = 20 / d (1) $1/\sqrt{3} = 20/d \Rightarrow d = 20\sqrt{3} = 20 \times 1{\cdot}73 = 34{\cdot}6$ m (1)	2
8	LHS = sin A/(1 + cos A) + (1 + cos A)/sin A = $[\sin^2 A + (1 + \cos A)^2] / [(1 + \cos A) \sin A]$ (1) = $[\sin^2 A + 1 + 2\cos A + \cos^2 A] / [(1 + \cos A) \sin A] = (2 + 2\cos A) / [(1 + \cos A) \sin A]$ (1) = $2(1 + \cos A) / [(1 + \cos A) \sin A] = 2/\sin A = 2 \operatorname{cosec} A$ = RHS (Hence proved). (1)	3
9	Data : O is the centre ; XY is a tangent at point P ; OP is the radius. (½) To Prove : OP ⊥ XY. Construction : Take any point Q (other than P) on XY and join OQ. (½) Proof : Q lies outside the circle (every point of a tangent except the point of contact lies outside), so OQ > OP. (1) This is true for every point Q on XY. Hence OP is the shortest distance from O to XY. The shortest distance from a point to a line is the perpendicular. ∴ OP ⊥ XY (Hence proved). (1)	3
10	Let tower CD = h ; from B (nearer) 60°, from A (farther) 30°, AB = 30 m. tan 60° = h/DB ⇒ DB = $h/\sqrt{3}$ (1) tan 30° = h/DA ⇒ DA = $h\sqrt{3}$; and DA – DB = AB = 30 (1) $h\sqrt{3} - h/\sqrt{3} = 30 \Rightarrow 2h/\sqrt{3} = 30$ (1) ∴ $h = 15\sqrt{3} = 15 \times 1{\cdot}73 \approx 25{\cdot}95$ m (1)	4

FA – 3 | ANSWER BOOKLET

Name : _____	**Roll No. : _____**
Marks Obtained : _____ / 20	**Invigilator's Signature : _____**

Q.No	Space for Answers

FORMATIVE ASSESSMENT

FA – 4

Statistics · Surface Areas and Volumes · Probability

In this section : Blue Print • Question Papers (Versions A, B, C, D) • Key Answers • Answer Booklet

FA – 4 | BLUE PRINT

Class X • Time : 45 Minutes • Maximum Marks : 20

Chapter	MCQ 1 m	SA 1 m	2 m	3 m	5 m	No. of Qns	Total Marks
Statistics	1 (Q3)	1 (Q5)	–	1 (Q9)	–	3	5
Surface Areas and Volumes	1 (Q1)	1 (Q4)	1 (Q7)	–	1 (Q10)	4	9
Probability	1 (Q2)	–	1 (Q6)	1 (Q8)	–	3	6
Total	**3×1=3**	**2×1=2**	**2×2=4**	**2×3=6**	**1×5=5**	**10**	**20**

Weightage to Objectives

Objective	Questions	Marks	Percentage
Knowledge	Q1, Q2, Q3, Q4, Q5	5	25%
Understanding	Q6, Q7	4	20%
Application / Skill	Q8, Q9, Q10	11	55%
Total	**10 Questions**	**20**	**100%**

School : _____

FORMATIVE ASSESSMENT — 4

Class X • Mathematics (81E) • Version A

Chapters : Statistics · Surface Areas and Volumes · Probability

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 3 × 1 = 3

1. The total surface area of a right circular cylinder having radius r and height h is .....
(A) πr(r + h) (B) 2πrh (C) 2πr(r – h) (D) 2πr(r + h)

2. For an event E, if P(E) = 0·75, then P(E) is .....
(A) 2·5 (B) 0·25 (C) 0·025 (D) 1·25

3. The empirical relationship between the three measures of central tendency is .....
(A) 3 Median = Mode + Mean (B) 3 Median = 2 Mode + Mean (C) 3 Median = 2 Mode + 2 Mean (D) 3 Median = Mode + 2 Mean

II. Answer the following questions. 2 × 1 = 2

4. Write the formula to find the total surface area of a solid hemisphere of radius r units.

5. Write the formula to find the mean of a grouped data by the direct method.

III. Answer the following questions. 2 × 2 = 4

6. Two dice whose faces are numbered from 1 to 6 are rolled simultaneously once. Find the probability that the sum of the numbers on their top faces is 10.

7. The radius of the base and the height of a right circular cylinder and a right circular cone are equal. If the volume of the cylinder is 360 cm^3, then find the volume of the cone.

IV. Answer the following questions. 2 × 3 = 6

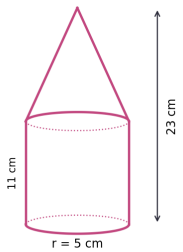
8. A bag contains 3 red balls, 5 white balls and 8 blue balls. One ball is taken out of the bag at random. Find the probability that the ball taken out is (a) a red ball (b) not a white ball.

9. Find the mean of the following frequency distribution by the direct method :

Class-interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	2	3	3	1	1

V. Answer the following question. 1 × 5 = 5

10. A solid is in the shape of a cone placed on a cylinder. The radii of both the cylinder and the cone are equal to 5 cm. If the height of the cylinder is 11 cm and the total height of the solid is 23 cm, find the curved surface area and the volume of the solid. (Take π = 22/7)



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School : _____

FORMATIVE ASSESSMENT — 4

Class X • Mathematics (81E) • Version B

Chapters : Statistics · Surface Areas and Volumes · Probability

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 3 × 1 = 3

1. The surface area of a sphere of radius 7 cm is .....
(A) 154 cm^2 (B) 308 cm^2 (C) 616 cm^2 (D) 1232 cm^2

2. The probability of a sure event is .....
(A) 1 (B) 0 (C) –1 (D) 1·5

3. In a data, the value of the observation that occurs most frequently is called the .....
(A) mean (B) mode (C) median (D) range

II. Answer the following questions. 2 × 1 = 2

4. Write the formula to find the curved surface area of a right circular cone of radius r units and slant height l units.

5. Write the empirical relationship between the mean, mode and median.

III. Answer the following questions. 2 × 2 = 4

6. A box contains cards numbered from 9 to 19. One card is drawn at random from the box. Find the probability that the card bears a prime number.

7. The radius of the base and the height of a right circular cylinder and a right circular cone are equal. If the volume of the cylinder is 27 cm^3, then find the volume of the cone.

IV. Answer the following questions. 2 × 3 = 6

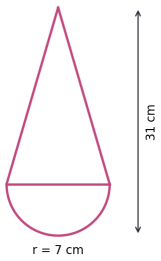
8. A box contains 6 red, 5 blue and 4 green balls. One ball is drawn at random from the box. Find the probability that the ball drawn is (a) not green (b) red.

9. Find the mean of the following frequency distribution by the direct method :

Class-interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	3	5	9	5	3

V. Answer the following question. 1 × 5 = 5

10. A wooden toy is made by mounting a cone on the circular base of a hemisphere. The radius of the hemisphere (and of the base of the cone) is 7 cm and the total height of the toy is 31 cm. Find the total surface area and the volume of the toy. (Take π = 22/7)



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School : _____

FORMATIVE ASSESSMENT — 4

Class X • Mathematics (81E) • Version C

Chapters : Statistics · Surface Areas and Volumes · Probability

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 3 × 1 = 3

1. The slant height of a cone whose base radius is 8 cm and height is 6 cm is .....
(A) 10 cm (B) 14 cm (C) 48 cm (D) 100 cm

2. If a fair die is rolled once, then the probability of getting an even number is .....
(A) 1/6 (B) 1/2 (C) 1/3 (D) 2/3

3. The abscissa of the point of intersection of the 'less than type' and 'more than type' ogives of a grouped data gives its .....
(A) mean (B) median (C) mode (D) range

II. Answer the following questions. 2 × 1 = 2

4. Write the formula to find the volume of a sphere of radius r units.

5. Write the empirical relationship between the mean, mode and median.

III. Answer the following questions. 2 × 2 = 4

6. A box contains 90 discs which are numbered from 1 to 90. One disc is drawn at random from the box. Find the probability that it bears a perfect square number.

7. A metallic sphere of radius 9 cm is melted and recast into the shape of a cylinder of radius 6 cm. Find the height of the cylinder.

IV. Answer the following questions. 2 × 3 = 6

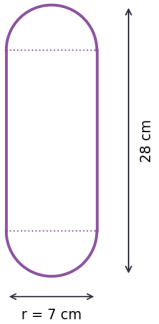
8. A bag contains 4 red, 6 white and 5 green balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is (a) red (b) not white.

9. Find the mean of the following frequency distribution by the direct method :

Class-interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	4	5	6	3	2

V. Answer the following question. 1 × 5 = 5

10. A solid is in the shape of a cylinder with two hemispheres of equal radii attached to its two ends. The radius is 7 cm and the total length of the solid is 28 cm. Find the total surface area and the volume of the solid. (Take π = 22/7)



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School : _____

FORMATIVE ASSESSMENT — 4

Class X • Mathematics (81E) • Version D

Chapters : Statistics · Surface Areas and Volumes · Probability

Time : 45 Minutes **Maximum Marks : 20**

I. Choose the correct alternative and write it with its letter. 3 × 1 = 3

1. The total surface area of a solid hemisphere of radius 7 cm is (π = 22/7) .....
(A) 308 cm^2 (B) 462 cm^2 (C) 616 cm^2 (D) 154 cm^2

2. The probability of an impossible event is .....
(A) 1 (B) 0 (C) –1 (D) 0·5

3. The median of the data 4, 6, 8, 10, 12 is .....
(A) 6 (B) 8 (C) 10 (D) 40

II. Answer the following questions. 2 × 1 = 2

4. Write the formula to find the volume of a right circular cone of base radius r units and height h units.

5. Write the empirical relationship between the mean, mode and median.

III. Answer the following questions. 2 × 2 = 4

6. A die is thrown once. Find the probability of getting (a) a prime number (b) a number greater than 4.

7. Find the volume of a right circular cone whose base radius is 7 cm and height is 6 cm. (Take π = 22/7)

IV. Answer the following questions. 2 × 3 = 6

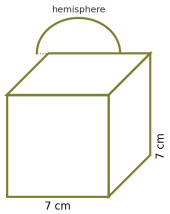
8. A bag contains 5 red, 4 white and 3 black balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is (a) white (b) not black.

9. Find the mean of the following frequency distribution by the direct method :

Class-interval	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency	6	5	4	3	2

V. Answer the following question. 1 × 5 = 5

10. A cubical block of side 7 cm is surmounted by a hemisphere of the largest possible diameter. Find the total surface area and the volume of the solid. (Take π = 22/7)



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FA – 4 | KEY ANSWERS • Version A

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(D) $2\pi r(r + h)$	1
2	(B) 0·25 [P(E) = 1 – P(E) = 1 – 0·75 = 0·25]	1
3	(D) 3 Median = Mode + 2 Mean	1
4	Total surface area of a solid hemisphere = $3\pi r^2$ sq.units	1
5	Mean $\bar{x} = \Sigma f_i x_i / \Sigma f_i$	1
6	Total outcomes n(S) = 36 (½) Favourable for sum 10 : {(4,6), (5,5), (6,4)} ⇒ n(A) = 3 (1) P = 3/36 = 1/12 (½)	2
7	Volume of cone = (1/3) × volume of cylinder [same radius and height] (1) = (1/3) × 360 = 120 cm^3 (1)	2
8	Total balls n(S) = 3 + 5 + 8 = 16 (1) (a) P(red) = 3/16 (1) (b) P(not white) = 1 – 5/16 = 11/16 (1)	3
9	C.I. 0–10, 10–20, 20–30, 30–40, 40–50 ; f_i 2, 3, 3, 1, 1 ; x_i 5, 15, 25, 35, 45 ; $f_i x_i$ 10, 45, 75, 35, 45 $\Sigma f_i = 10$; $\Sigma f_i x_i = 210$ (1½) Mean = $\Sigma f_i x_i / \Sigma f_i = 210 / 10$ (1) ∴ Mean = 21 (½)	3
10	r = 5 cm, cylinder height h = 11 cm, cone height $h_1 = 23 – 11 = 12$ cm (½) Slant height $l = \sqrt{(h_1^2 + r^2)} = \sqrt{(144 + 25)} = \sqrt{169} = 13$ cm (1) CSA = $\pi rl + 2\pi rh = \pi r(l + 2h) = (22/7) \times 5 \times (13 + 22) = (22/7) \times 5 \times 35 = 550$ cm^2 (1½) Volume = $\pi r^2 h + (1/3)\pi r^2 h_1 = \pi r^2(h + h_1/3) = (22/7) \times 25 \times (11 + 4) = (22/7) \times 25 \times 15 \approx 1178{\cdot}57$ cm^3 (2)	5

FA – 4 | KEY ANSWERS • Version B

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(C) 616 cm^2 [$4\pi r^2 = 4 \times (22/7) \times 7 \times 7 = 616\ cm^2$]	1
2	(A) 1	1
3	(B) mode	1
4	Curved surface area of a cone = πrl sq.units	1
5	3 Median = Mode + 2 Mean (empirical relationship between the three measures of central tendency)	1
6	Cards 9 to 19 ⇒ total n(S) = 11 (½) Prime numbers : 11, 13, 17, 19 ⇒ n(A) = 4 (1) P(prime) = 4/11 (½)	2
7	Volume of cone = (1/3) × volume of cylinder [equal radius and height] (1) = (1/3) × 27 = 9 cm^3 (1)	2
8	Total balls n(S) = 6 + 5 + 4 = 15 (1) (a) P(not green) = 1 – P(green) = 1 – 4/15 = 11/15 (1) (b) P(red) = 6/15 = 2/5 (1)	3
9	C.I. 0–10, 10–20, 20–30, 30–40, 40–50 ; f_i 3, 5, 9, 5, 3 ; x_i 5, 15, 25, 35, 45 ; f_ix_i 15, 75, 225, 175, 135 $\Sigma f_i = 25$; $\Sigma f_ix_i = 625$ (1½) Mean = $\Sigma f_ix_i / \Sigma f_i = 625 / 25$ (1) ∴ Mean = 25 (½)	3
10	r = 7 cm ; total height = 31 cm ⇒ cone height h = 31 – 7 = 24 cm (½) Slant height l = $\sqrt{(h^2 + r^2)} = \sqrt{(576 + 49)} = \sqrt{625} = 25$ cm (1) TSA = CSA of cone + CSA of hemisphere = $\pi rl + 2\pi r^2 = \pi r(l + 2r)$ (1) = (22/7) × 7 × (25 + 14) = 22 × 39 = 858 cm^2 (1) Volume = $(1/3)\pi r^2 h + (2/3)\pi r^3 = (1/3)\pi r^2(h + 2r)$ (½) = (1/3) × (22/7) × 49 × (24 + 14) = (1/3) × 154 × 38 ≈ 1950·67 cm^3 (1)	5

FA – 4 | KEY ANSWERS • Version C

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(A) 10 cm [$l = \sqrt{(r^2 + h^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$ cm]	1
2	(B) 1/2 [even numbers {2, 4, 6} ⇒ P = 3/6 = 1/2]	1
3	(B) median	1
4	Volume of a sphere = (4/3) πr^3 cubic units	1
5	3 Median = Mode + 2 Mean (empirical relationship between the three measures of central tendency)	1
6	Total discs n(S) = 90 (½) Perfect squares from 1 to 90 : 1, 4, 9, 16, 25, 36, 49, 64, 81 ⇒ n(A) = 9 (1) P(perfect square) = 9/90 = 1/10 (½)	2
7	Volume of cylinder = Volume of sphere (1) $\pi \times 6^2 \times h = (4/3)\,\pi \times 9^3 \Rightarrow 36h = (4/3) \times 729 = 972 \Rightarrow h = 27$ cm (1)	2
8	Total balls n(S) = 4 + 6 + 5 = 15 (1) (a) P(red) = 4/15 (1) (b) P(not white) = 1 – P(white) = 1 – 6/15 = 9/15 = 3/5 (1)	3
9	C.I. 0–10, 10–20, 20–30, 30–40, 40–50 ; f_i 4, 5, 6, 3, 2 ; x_i 5, 15, 25, 35, 45 ; f_ix_i 20, 75, 150, 105, 90 $\Sigma f_i = 20$; $\Sigma f_ix_i = 440$ (1½) Mean = $\Sigma f_ix_i / \Sigma f_i = 440 / 20$ (1) ∴ Mean = 22 (½)	3
10	r = 7 cm ; total length = 28 cm ⇒ cylinder height h = 28 – 2(7) = 14 cm (½) TSA = CSA of cylinder + 2 × CSA of hemisphere = $2\pi rh + 2(2\pi r^2) = 2\pi r(h + 2r)$ (1) $= 2 \times (22/7) \times 7 \times (14 + 14) = 44 \times 28 = 1232$ cm^2 (1) Volume = $\pi r^2h + 2(2/3)\pi r^3 = \pi r^2h + (4/3)\pi r^3$ (½) $= (22/7)\times49\times14 + (4/3)(22/7)\times343 = 2156 + 1437{\cdot}33$ (1) ∴ Volume ≈ 3593·33 cm^3 (1)	5

FA – 4 | KEY ANSWERS • Version D

Class X • Mathematics (81E) • Maximum Marks : 20

Q.	Value Points	Marks
1	(B) 462 cm^2 [TSA of solid hemisphere = $3\pi r^2 = 3 \times (22/7) \times 7 \times 7 = 462\ cm^2$]	1
2	(B) 0	1
3	(B) 8 [the middle value of 4, 6, 8, 10, 12]	1
4	Volume of a cone = (1/3) $\pi r^2 h$ cubic units	1
5	3 Median = Mode + 2 Mean (empirical relationship between the three measures of central tendency)	1
6	Total outcomes n(S) = 6 (a) prime numbers {2, 3, 5} ⇒ P(prime) = 3/6 = 1/2 (1) (b) numbers greater than 4 : {5, 6} ⇒ P = 2/6 = 1/3 (1)	2
7	Volume of cone = (1/3)$\pi r^2 h$ (1) = (1/3) × (22/7) × 7^2 × 6 = (1/3) × 22 × 7 × 6 = 308 cm^3 (1)	2
8	Total balls n(S) = 5 + 4 + 3 = 12 (1) (a) P(white) = 4/12 = 1/3 (1) (b) P(not black) = 1 – P(black) = 1 – 3/12 = 9/12 = 3/4 (1)	3
9	C.I. 0–10, 10–20, 20–30, 30–40, 40–50 ; f_i 6, 5, 4, 3, 2 ; x_i 5, 15, 25, 35, 45 ; $f_i x_i$ 30, 75, 100, 105, 90 $\Sigma f_i = 20$; $\Sigma f_i x_i = 400$ (1½) Mean = $\Sigma f_i x_i / \Sigma f_i = 400 / 20$ (1) ∴ Mean = 20 (½)	3
10	Largest hemisphere diameter = side = 7 cm ⇒ radius r = 3·5 cm. (½) TSA = (surface area of cube) – (base circle) + (CSA of hemisphere) = $6a^2 - \pi r^2 + 2\pi r^2$ = $6a^2 + \pi r^2$ (1) = 6 × 7^2 + (22/7) × 3·5 × 3·5 = 294 + 38·5 = 332·5 cm^2 (1) Volume = (volume of cube) + (volume of hemisphere) = $a^3 + (2/3)\pi r^3$ (1) = 7^3 + (2/3) × (22/7) × $(3·5)^3$ = 343 + 89·83 (1) ∴ Volume ≈ 432·83 cm^3 (½)	5

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Name : _____	**Roll No. : _____**
Marks Obtained : _____ / 20	**Invigilator's Signature : _____**

Q.No	Space for Answers